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Advancements in Weather Forecasting and Climate Modeling Technology

Abstract

Traditional methods of weather forecasting include the use of numerical weather prediction (NWP). Yet, evidence continues to suggest that machine-learning (ML) models surpass NWP models on multiple levels. Thus, many of the recent developments in improving forecast accuracy employ various strategies of ML technology through the data assimilation, processing, and analysis of real-time and historical forecast data. Strategies proposed and analyzed in this review include radar systems, data-driven frameworks, and a vast number of ML methods. Though many of these methods were proposed, in part, to minimize the current drawbacks of forecasting technology, there remain shortcomings – inhibiting prediction accuracy – which the technology must address. While reflecting generally enhanced accuracy to previous methods, the reviewed technologies also tend to enable reduced computational costs and processing speed, making them a favorable option to pursue. Overall, machine-learning techniques applied mainly to the processing and modeling of various recorded aspects of climate and weather data have produced significantly improved results over conventional, often NWP, methods of forecasting.

Introduction

Meteorology is an evolving science and weather forecasting a valuable tool, especially as climate change causes more erratic [1] and extreme weather, making forecasting of greater necessity and difficulty. Weather forecasting has conventionally relied on systems of numerical weather prediction (NWP) which integrate data assimilation (DA), physics-based models, and global observations. Attempts at further advancement in forecasting technology have been met with challenges in resolution, computational costs, and the underuse of vast observational datasets [2]. Advances in weather prediction methods include satellite and remote sensing [1] technology as well as machine-learning (ML) and algorithm-based systems – which are often dependent upon satellite or multi-satellite data. To enhance the capabilities and accuracy of weather forecasting, distinctions have been made beyond technology and algorithm type to encompass specific weather conditions, levels of precipitation, and methods of analysis and processing. This review aims to address and describe some of the recently published advances in weather forecasting systems, while also evaluating the different phases which lead to the composition of these forecasts.

Methods

Reference Criteria

The references cited in this review were obtained from databases including *Nature*, *Web of Science*, *EBSCO* and *Green FILE*. Databases within EBSCO were further filtered to include only considerably reputable sources. The search criteria entered into the databases was “forecasting AND weather OR climate”. Articles which employ weather and climate modelling technology for other purposes were excluded, so as not to distract from this review’s main subject. Each article was published either in 2025 or 2026. Thus, the proposed methods are relatively up-to-date with currently explored technologies in this field. Most articles propose and report on a specific method; however, one of the references for this review is a review article regarding the integration of Artificial Intelligence (AI) into extreme weather and climate forecasting.

Review of Methods

Various technologies, computational strategies, data and weather types have been leveraged to advance the methods of weather and climate forecasting and modeling. Each proposed method is an enhancement of previous attempts to improve forecasting but come with their own respective weaknesses – such as size of dataset and range of prediction – as well as strengths – including accuracy, cost, and processing speed. Key distinctions in the capabilities of each method of forecasting can be determined at the levels of data collection, processing, modeling and analysis (See Figure 1).

Figure 1. Analysis of 8 Different Forecasting Methods

Method	Purpose	Data	Function & Process	Analysis & Modeling	Source
Radar systems in satellite instrumentation technology	Short-Term Weather Prediction and Event Warnings Despite significant advancements in signal and image processing, employed in satellite imagery methodologies, the technology continues to lack capabilities for practical use cases. Thus, newer satellite platforms aim to address this shortcoming.	The Automatic Picture Transmission (APT) TV Camera System operates on the the Low Earth Orbit (LEO) meteorological satellite. The camera provides global coverage of cloud formations on Earth, collecting daytime and nighttime cloud formation imagery. The satellite uses the advantages of low-Earth-orbit-based satellite constellation deployments as a means of near-real-time tracking of precipitation profiles and atmospheric hydrometers. Through Doppler and range processing, radar systems create backscatter images for detecting snow, hail, rain, ocean surface wave height, wind, frequency and direction, clouds, wind speed, vertical precipitation profiles, temperature, and atmospheric moisture content.	Raw data is processed by cloud-based analytics tools, and further fed into advanced forecasting models which use Artificial Neural Networks (ANNs) designed for remote sensing. These models must operate in near-real-time to allow for timely production of extreme weather event warnings and other such essential outputs. Techniques employed include statistical regression for deriving statistical relationships amongst atmospheric parameters. Also used are linear regression methods to compute a linear fit which minimizes the sum of squared errors between observed and model data. However, nonlinear regression methods, such as ANNs, have been proven better suited for effective forecasting and capturing of relationships in remote sensing datasets.	The method of satellite instrumentation technology combines collected data into big-data lakes on internet cloud platforms, then, for remote sensing, constructs AI-based multi-layered weather prediction models.	[1]
FuXi Weather	ML-Based Global Forecasting System There are continued disparities between the resources, and thus accuracy, contributing to forecasting in wealthier versus low-income regions. Hence the necessity of a global forecasting system. Further, NWP models require increased computational costs, while ML offers a promising, less costly, alternative.	FuXi assimilates multi-satellite data, collected by the Global Navigation Satellite System (GNSS) and three polar-orbiting meteorological satellites. Missing data is accounted for through a masking technique which assigns a 1 for available data, otherwise assigning a 0.	Differing surface and upper-air variables are processed separately, with a refinement module applied, for improved accuracy. Global weather forecasts are generated every six hours by employing satellite data preprocessing, DA, – via FuXi-DA.	Forecasting is enacted through the FuXi-Short model, then refined by FuXi-DA analysis fields to reconcile differences in accuracy. Forecasts are analyzed against benchmark datasets at corresponding forecast times.	[2]
Elastic Net Regularization (EN), XGBoosting (XGB), Random Forest (RF), Support Vector Machine (SVM), Multiple Linear Regression (MLR) 5 ML methods	Climate Forecasting These ML methods are intended to forecast air temperature and precipitation in different climate regions across Brazil. Climate forecasting is tied to teleconnection patterns, which are natural phenomena occurring at certain locations and impacting climate and its variability in more remote areas. Additionally, multiple teleconnection patterns may occur at once, making prediction of greater difficulty.	Data was taken between 1979 and 2022, then split between training (34 years) phase and testing (10 years) phase. The dataset contains monthly data on accumulated precipitation (mm) and mean air temperature (°C) acquired from the database of conventional and automatic weather stations, under the purview of the Brazilian National Institute of Meteorology. Missing values were filled mainly by use of gridded datasets, based on spatial correlation between station-based observations and gridded data.	Some methods of data processing involve locating outliers and considering them through visual inspection to minimize impact on the analysis. Seasonal dummies and lags in the dependent variables were incorporated into the data processing to account for seasonal variations in climate. EN is a kind of regularized regression. XGB is a general form of boosting to minimize cost function and introduce randomness into the algorithm. RF is a ML technique, which averages the results of decision trees. SVM classifies a data set through multi-dimensional feature mapping. MLR produces a linear model with multiple independent predictor variables.	Models were verified using the correlation coefficient, mean error, coefficient of determination, root mean square error, index of agreement, performance index, and percentage BIAS index. To identify possible fit failures amongst the models, diagnostic and stationarity tests were applied.	[3]
Novel historical feature reuse scheme of PV	Numerical Weather Forecasting Meant to enhance the accuracy of the PV power forecasting model.	Historical data is acquired as a research object from a photovoltaic power station set up in Jilin Province, China. Measurements included in the dataset are irradiance intensity, meteorological forecast and power generation data.	First proposed was a WT classification method based on the Elkan K-means algorithm, along with a feature matching mechanism based on the Markov distance to merge information. A bidirectional recurrent residual network was then constructed to improve feature extraction performance for differing PV output scenarios of the forecasting model.	The PV reuse scheme recommends an error decoupling mechanism to evaluate the model's upper limit of forecasting accuracy	[4]
European Centre for Medium-Range Weather Forecasts (ECMWF), Community Earth System Model, version 2 (CESM2) Two seasonal hindcast products	Seasonal Climate Forecasts Focuses on precipitation-relevant weather types (WTs) in the United States. The predictors used to classify weather types are sea level pressure, uv winds and water vapor mixing ratio at 700 hPa.	The operational model, ECMWF, which is initialized each month, was developed using numerical weather prediction (NWP). The research model, CESM2, is initialized on a quarterly basis and generates the Seasonal-to-Multisyear Large Ensemble (SMYLE), which contains 20 ensemble members. The ECMWF and CESM2 were validated from 1993 to 2019 for analysis, with twelve WTs and four seasons represented in the data which the algorithm processes and clusters.	A 31-day moving average Gaussian filter is imposed upon predictor fields. Then, filtered data are deleted from the original data to construct predictor anomalies as a means of retaining weather variability while removing variability of longer time scales. Predictor anomalies are standardized by daily standard deviation and regional mean. Such a standardization enables a more even assessment of each predictor's significance in clustering.	Seasonal forecasts of WTs, such as recurring large-scale atmospheric patterns, are analyzed using a clustering method of large-scale predictors to express precipitation variability across the United States. A k-means clustering method is applied with the initial seed from the output of hierarchical clustering.	[5]
Data-driven short-term photovoltaic (PV) forecasting framework on the basis of the gated recurrent units (GRU) and Kolmogorov-Arnold networks (KAN)	Forecasting under Extreme Weather Conditions As a result of increased occurrences of extreme weather events, PV generation can be substantially reduced, threatening power system stability. This is a method of addressing this issue, thereby enhancing forecast accuracy.	Historical PV data are preprocessed, with key meteorological factors extracted via Grey Relational Analysis (GRA) – to measure the degree of relationship amongst sequences.	Normal weather conditions are clustered into categories of overcast, cloudy, and sunny using K-Means++. Then, a data-driven method identifies extreme weather conditions characterized by limited data before a similar-day selection method is leveraged to construct high-quality training sets.	A GRU-KAN model, integrating nonlinear approximation power and temporal modeling capability, is proposed for PV forecasting under extreme and normal conditions. Then, to enhance forecasting performance for cases of limited data, the neuron structure of the feature extraction layer is simplified	[6]
Aardvark Weather	Aardvark Weather is an end-to-end data-driven system for weather prediction. This system is able to generate predictions without the input of traditional NWP, but rather by learning a mapping from raw input observations. ML transforms NWP by utilizing neural networks in place of the numerical solver. Such a transformation will improve accuracy and speed of the forecasting component of the prediction pipeline, as current models depend on numerical systems at initialization and to develop local forecasts, thus restraining the achievable gains.	Aardvark creates forecasts of specific humidity, northward and eastward wind, geopotential and temperature – amongst other factors. Several in situ and remote sensing observations are selected as inputs for the atmospheric state estimation model. To certify that an NWP system is not necessary for operational development, data is selected only at a processing level of 1B or 1C. Level 1B satellite data are calibrated and geolocated, have processed raw sensor measurements, corrections for instrument and sensor biases, but are still in the form of physical measurements. Whereas level 1C satellite data is further processed to include geometric and radiometric corrections.	Aardvark consists of three modules and leverages high-quality reanalysis data while training and also being entirely independent of NWP products at the time of deployment. The modules include an encoder, processor, and decoder. The encoder obtains observational data both on and off the grid, then produces a gridded initial state. When the initial atmospheric state is estimated, it is inputted into the processor module which creates a gridded forecast at lead time of twenty-four hours. Lastly, the decoders intake the forecasts and output local predictions.	Aardvark ingests observations, then produces local station and global gridded forecasts. Aardvark is a neural process model, using a group of deep learning models which provide a malleable framework able to learn from off grid, sparse, and missing data. Probabilistic predictions for arbitrary regions are then provided and ideally fit for complex environmental data.	[7]
Free space optics (FSO) operating along high-speed trains with broadband service	Rapidly growing over current radio frequency technology, FSO may be easily implemented and potentially overcome the bandwidth constraint problem of high-speed trains – especially in dynamic environments.	Implemented under data of atmospheric weather forecast conditions.	To implement this method, there was a formulation model and mathematical derivation for double-curved track base-to-train communications (B2T) based FSO link, as well as a proposed multiple transmitters configuration to maximize coverage range and hone the B2T – FSO link performance under various weather conditions, such as levels of rain and fog attenuation.	Meanwhile, the double curve track B2T – FSO model was evaluated in terms of signal-to-noise ratio (SNR), eye diagrams, received power, and bit error rate (BER).	[8]

While a significant portion of the data employed by weather forecasting systems is drawn from satellites, it is valuable to understand the specific methods of data collection which populate a database. Some factors may be considered independently or as a contributor to a larger process or weather event. As the methods of forecasting differ, so too must the techniques for data processing, making processing an invaluable step in modeling. The input must be fitted to the

technology or algorithm being used and catered to the desired output. Lastly, the methods of modeling and analyzing input are important aspects of interpreting data. More significantly, these methods lead to enhanced forecasting capabilities by overlaying various contributors to the weather. Different approaches, such as Multiple Linear Regression [3] and error decoupling [4] also reflect differences in data, technology, and process technique.

Discussion

Machine-learning [2,5] and photovoltaic (PV) [6,4] methods are suggested as having great potential for supporting traditional climate models, particularly NWP systems, by bridging current gaps – including challenges in comprehending and quantifying the actual effects of teleconnection patterns, as well as response time in which effects on climate are observed. In contrast, other technologies [7] introduce methods that should replace or function without use of NWP systems.

For the seasonal hindcast products [5], the spatial precipitation patterns corresponding to each weather type (WT) are captured less well in the forecasting systems than are the large-scale variables from which the WTs are being derived. However, the WT patterns themselves are represented well in the European Centre for Medium-Range Weather Forecasts (ECMWF) and Community Earth System Model, version 2 (CESM2) seasonal forecasting systems – with ECMWF predictions being somewhat closer to observed weather data. The forecasted WT frequency results assert that both forecasting systems are similarly precise, and the greatest discrepancies arise depending on the WT and season. Specifically, summer WT frequencies tend to be less predictable than winter. Both prediction systems underestimate the interannual frequency spread, – potentially a result of ensemble averaging – yet do well in capturing frequency rank order. Analysis has shown that forecasting of climate teleconnection indices alone would be insufficient for representing precipitation variability associated with the WTs. Comparable results from two initialized prediction models, originating from different sides of the operations-research and weather-climate spectrum, are promising and contribute to multi-model initialized prediction efforts and WT forecasting [5]. However, FuXi Weather [2] outperforms the ECMWF high-resolution forecasts in observation-sparse regions and beyond day one of

prediction – and specifically at lead times of 2.00, 2.25, and 3.25 days for 300, 500, 850 hPa. FuXi generates reliable 10-day forecasting, also using fewer observations than conventional NWP systems.

The global forecasts generated by Aardvark Weather [7] outperform operation NWP baseline, as well as state-of-the-art end-to-end forecasting systems containing input from human forecasters. End-to-end tuning further improves the accuracy of local forecasts, and results in evaluating station and global forecasting, and analyzing encoder module ablation, demonstrate skillful forecasting is achievable without depending on NWP at deployment time. This will allow the realization of full accuracy and speed benefits of data-driven models. Overall, Aardvark Weather is capable of producing affordable, rapid development of customized models, while also enabling reduced computational costs.

Experimental results of the data-driven short-term PV forecasting framework [6] demonstrate the model reaching the highest average coefficient of determination at 96.90% and lowest average Root Mean Square Error (RMSE) of 3.86 under normal conditions. The model has also achieved the lowest RMSEs – 0.900, 0.950, 1.157, 1.438 – across four days representing extreme weather. With the application of the reuse scheme [4] to the PV model, data offered by the PV power station in Jilin Province, China was extracted as a research object. The results produced then verified the validity of this proposed model with day-ahead power forecasting accuracy reaching 91.12%.

Additionally, results have been analyzed to show an improvement in high-speed trains with broadband service, and that the B2T-FSO link [8] can be implemented under the impact of weather conditions. However, results vary between analyses of SNR levels, received power, BER and eye diagrams.

According to statistical indices, the five ML techniques [3] – Elastic Net Regularization (EN), XGBoosting (XGB), Random Forest (RF), Support Vector Machine (SVM), Multiple Linear Regression (MLR) – performed well, overall, during training phases, especially the RF and XGB techniques. The testing phase showed MLR models outperforming others in estimating temperature, though no consistent pattern emerged with regard to the best method of estimating precipitation across regions. The results are expressed as exhibiting great potential for employing

diverse ML techniques to estimate meteorological variables in areas with varying climate characteristics.

Outlook

Machine-learning methods have been a valuable resource in a variety of fields – with many applications being proposed and tested – but with issues still unresolved and more potential uses unexplored. The application of ML to prediction methods such as weather forecasts and climate models has been particularly fruitful in comparison to prior techniques. Yet, the methods in this review appear to focus significantly on the analysis of precipitation levels. Though other factors are considered – differing amongst methods – it would be beneficial to experiment with these technologies under a wider variety of conditions. Further, it would be beneficial to examine the effect of inputting conditions which have been employed by other technologies. An array of factors impact weather and climate in subtle and overt ways; thus, it is important that models are able to handle the interaction of a more diverse set of data.

A particularly important consideration is that, with the increased occurrence of natural disasters, and the value of predicting them, models are tailored to the forecasting of extreme weather events. Though this may be a long-term goal for each of the reviewed methods, variability is inherent even in less extreme natural phenomena and should therefore be a continued consideration as models are constructed.

Conclusions

It has been asserted that a single ML model may replace an entire NWP pipeline [7]. Enhancing the accuracy of weather and climate forecasting, as well as communication of extreme events, is of greater necessity. Therefore, new technologies, algorithms, and uses for predictive systems are being developed and analyzed. Such technologies include the use of remote sensing and radar systems through satellite instrumentation technology. ML is an especially well-documented, emerging advancement, given these systems are computationally cheaper, quicker, and easier to

maintain and improve. So, various algorithms and refined datasets are being implemented for forecasting systems. Such datasets have often been enhanced so that distinct weather conditions may be more easily identified and analyzed, thus enhancing the accuracy of predicting particular weather events. In fact, forecasting based on weather types associated with varying levels of precipitation is proposed [5] with the intent to complement existing systems for seasonal prediction.

Results have proven each method promising at various aspects of forecasting, particularly against conventional forecasting models. Further, some of the systems detailed in this paper build upon, and further surpass, others, in terms of recent analyses of the results. There remain challenges in the new systems, as seen in past ones, including limited observation subsets [2]; model overfitting [3]; reduced forecast activity [2]; and, in some cases, scalability [8]. However, feasible solutions have been presented to eventually address many remaining challenges, and the accuracy of the systems is still superior to prior implementations. Additionally, there are promising outlooks for this technology in other fields, even on a grander scale.

Ethics Declaration

I previously used artificial intelligence as a means of presenting and enhancing weather forecasts by creating an AI agent. However, I do not believe I possess any favor toward Machine Learning methods of prediction over alternate, promising methods and technologies.

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Technology Disclosure

I used a laptop for the following applications and websites:

Google, NULibrary, EBSCO and GreenFILE, Nature, Web of Science, Microsoft Word

I used a phone for the following applications:

Microsoft Word, Google, Nature Communications

References

Articles / Journals

1. Nardi, E.; Cohen, O.; Pinhasi, Y.; Haridim, M.; & Gavan, J. (2025). Weather Forecasting Satellites—Past, Present, & Future. *Information*, 16(8), 677. <https://doi.org/10.3390/info16080677>
2. Sun, X.; Zhong, X.; Xu, X.; et al. (2025) A data-to-forecast machine learning system for global weather. *Nature Communications* 16, 6658. <https://doi.org/10.1038/s41467-025-62024-1>
3. Henrique de Freitas, C.; Kaore Lago Kitagawa, Y.; Barros Ferreira, D.; et al. (2025). Use of Machine Learning Techniques for Monthly Precipitation and Air Temperature Forecasting in Different Climate Regions Across Brazil. *International Journal of Climatology*, 45(14), 70098. <https://doi.org/10.1002/joc.70098>
4. Yang, M.; Guo, Z.; Wang, D.; et al. (2026). Short-term photovoltaic power forecasting method considering historical information reuse and numerical weather forecasting. *Renewable Energy*, 256, 123933. <https://doi.org/10.1016/j.renene.2025.123933>
5. Towler, E.; Done, J.M.; Ge, M.; Gilleland, E.; & Prein, A. F. (202). Seasonal Forecasting of Precipitation-Relevant Weather Types over the United States. *Weather & Forecasting*, 40(11), 2239–2253. <https://doi.org/10.1175/WAF-D-24-0230.1>
6. Lyu, J.; Zhu, G.; He, C. (2026). Data-driven short-term photovoltaic power forecasting under extreme weather conditions using GRU-KAN model. *Renewable Energy*, 257, 124756. <https://doi.org/10.1016/j.renene.2025.124756>
7. Allen, A.; Markou, S.; Tebbutt, W.; et al. (2025). End-to-end data-driven weather prediction. *Nature* 641, 1172–1179. <https://doi.org/10.1038/s41586-025-08897-0>
8. Alzahmi, A.; Abdullah, M.F.L.; Alhartomi, M.A.; et al. (2025). FSO mathematical modelling and optimization base track to train communication under the atmospheric weather forecast conditions. *Scientific Reports* 15, 15552. <https://doi.org/10.1038/s41598-025-97655-3>

Review

- Camps-Valls, G.; Fernández-Torres, M.Á.; et al. (2025). Artificial intelligence for modeling and understanding extreme weather and climate events. *Nature Communications* **16**, 1919. <https://doi.org/10.1038/s41467-025-56573-8>